Stretch 2 Component Test Runbook

Pill bottle project · bottle_tracking package · T0, T1a, T1b, T2, T3, T4, T6, T8, T9, T10, T11

> Run everything on the robot over `ssh -Y hello-robot@192.168.1.172`. Type commands one per line. Don't add `#` comments on the same line in your Mac's terminal (zsh treats them as part of the command).

auto tester logs results by itself manual log each trial with run_manual_test.py

One time only

1. Install YOLO and find your model

```
pip3 install ultralytics "numpy<2"
find ~ -name "*.pt" 2>/dev/null
```

Write down the full path of your trained `best.pt`. Keep `numpy<2`, because NumPy 2 breaks ROS 2 Humble.

2. Rebuild after pulling the update

```
cd ~/stretch2pillbottlesort
git fetch
git reset --hard origin/main
cd ~/ament_ws
colcon build --packages-select bottle_tracking --symlink-install
source install/setup.bash
```

3. Check the detector sees the bottle (upright image, green box)

Start the session terminals below first, then run:

```
ros2 run rqt_image_view rqt_image_view /pill_bottle/debug_image
```

If the image is sideways, set `rotate_image` in `ros2_ws/src/bottle_tracking/config/params.yaml` to `cw`, `ccw` or `none` (whichever matches your training images), then restart terminal 3. A red box means it sees the bottle but has no depth there.

4. Calibrate the arm (before T1a and T1b)

The arm and wrist will move, so clear the robot's right side first.

```
ros2 run bottle_tracking calibrate_arm --ros-args -p write:=true
```

Then put your pick code's gripper values into `gripper_open` and `gripper_closed` in params.yaml.

Every session: setup

If the robot was powered off, home it first, with the driver stopped:

```
stretch_robot_home.py
stretch_robot_stow.py
```

Open 4 terminals and run `source ~/ament_ws/install/setup.bash` in each one first.

Terminal	Command (leave running)
1	`ros2 launch stretch_core stretch_driver.launch.py broadcast_odom_tf:=True mode:=navigation`
2	`ros2 launch stretch_core d435i_high_resolution.launch.py`
3	`ros2 launch bottle_tracking bottle_tracking.launch.py model_path:=/full/path/to/best.pt`
4	`python3 ~/stretch2pillbottlesort/testing/watch_bottle.py` (prints SEES / LOST)

In a 5th terminal for the tests:

```
cd ~/stretch2pillbottlesort
P=$(ros2 pkg prefix bottle_tracking)/share/bottle_tracking/config/params.yaml
```

Every tester pauses between trials so you can move the bottle. Press Enter to skip the wait, or change it with `-p pause_s:=30`.

Tests

T0: Logging setup

```
python3 testing/summarize_results.py
```

Paste the output into the T0 issue and close it.

T1a: Find the bottle and print the plan (no motion) auto

Searches with the head, then prints the base moves and the lift/arm/gripper steps. The base and arm don't move. Afterward it asks whether the plan looks right.

```
ros2 run bottle_tracking t1a_plan_tester --ros-args --params-file $P -p trials:=3
```

T1b: Find, plan, drive and grab auto

Same as T1a, then it drives (Enter to confirm), looks again up close, re-plans the arm, grabs (Enter to confirm), lifts and holds for 5 s. You answer what happened, then it puts the bottle back. Stow the arm and clear the path first.

```
ros2 run bottle_tracking t1b_execute_tester --ros-args --params-file $P -p trials:=3
```

T2: Find the bottle and say where it is auto

Uses the head only (the base never moves). Reports whether it sees the bottle, its position in the fixed frame, its distance and direction, and whether the robot needs to move to grab it. 20 s pause between trials.

```
ros2 run bottle_tracking t2_find_tester --ros-args --params-file $P -p trials:=5
```

T3: Watch while moving, then remember auto

Part A: moves around with head tracking on (turns, drives, a mix). Part B runs right after: it finds the bottle first (turning the base if needed), then moves with the head frozen and looks back at the remembered spot. Leave about 1 m of clear space.

```
ros2 run bottle_tracking t3_tracking_tester --ros-args --params-file $P -p trials:=3
```

To run only one part: `-p part:=A` or `-p part:=B`

T4: "Nothing there" recovery manual

Let the robot find the bottle, then take the bottle away. The watcher should print LOST, and this should show `MISSING` within about 3 s:

```
ros2 topic echo /bottle_memory/state --once
python3 testing/run_manual_test.py T4
```

T6: Distance at different spots, base still auto

60 s pause between spots. Prints the distance from the base and from the camera, and how steady the reading is. Type the tape-measured distance (or press Enter to skip) to get the error.

```
ros2 run bottle_tracking t6_distance_tester --ros-args --params-file $P -p trials:=5
```

T8: Grasp only manual

Put the gripper around the bottle, close, lift and hold for 5 s:

```
python3 testing/run_manual_test.py T8
```

T9, T10, T11: Orientation, in-hand verify, retry manual

```
python3 testing/run_manual_test.py T9
python3 testing/run_manual_test.py T10
python3 testing/run_manual_test.py T11
```

Manual tester keys: `s` SUCCESS, `m` MISS, `n` NOT_FOUND, `d` DROP, `t` TIMEOUT, `f` FALSE_POSITIVE, `e` ERROR, `q` quit.

End of every session

```
python3 testing/summarize_results.py --session latest
python3 testing/summarize_results.py --test T2 --markdown
```

Paste each test's markdown into its GitHub issue.

If something doesn't work

Problem	Fix
Testers never find the bottle	Check the debug image. Lower `conf_threshold` or fix `rotate_image` in params.yaml.
Head turns away from the bottle	Set `pan_sign: -1.0` (or `tilt_sign`) in params.yaml and restart terminal 3.
Arm stops short or goes too far	Run `calibrate_arm` again.
Gripper doesn't close enough	Lower `gripper_closed` (check with T8).
Distances are always off by the same amount	Change `depth_offset_m` by the average signed error from T6.